EFY Note
The source code of this project is available for free download at *source.efymag.com*

If you want the same amount of liquid again, you just swipe the RFID card again across the RFID reader. You can repeat this again and again till the amount of liquid in the container finishes.

Software

The software code (liquid_vending.ino) is written in Arduino programming language. Arduino IDE is used for compiling and uploading the sketch to ATmega328 microcontroller of the Arduino board. Every RFID tag has a unique number. This number has to be included in the Arduino code/sketch.

Arduino code uses LiquidCrystal.h header file for the LCD and serial communication for reading the data from the RFID reader. Two built-in functions are used, namely, Serial.available( ) to return the RFID tag number arrived in the serial buffer and Serial.readString( ) to read a string of RFID tag numbers.

In this code/sketch, we used a variable as 'int amount = 1000'. This gives a default liquid amount of 1000 millilitres or one litre per card swipe. Just change the amount value as per your requirement.

Construction and testing

1. Connect all the components and modules as per the schematic diagram.

2. Connect a 12V power supply from 12V adaptor or 12V battery to Vin pin of Arduino, relay driver and to relay NO pin.

3. Connect relay common pin to the solenoid as shown in the circuit diagram.

4. Check the default amount of liquid in the code for dispensing per minute.

5. Swipe the RFID tag over EM-18 reader. If it matches the programmed RFID tag code, relay and solenoid will be energised, water from the source(tap) will flow through flow sensor.

6. You will get all relevant information on LCD1 as already explained.

Make sure that there is sufficient amount of liquid in the source (water tap or container). Less amount of liquid in the source means less pressure in liquid flow in the pipe, resulting in inaccuracy in the liquid flow measurement. EFY


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